EE 66 Fall 2026 — Signals, Dynamics, and Information — Discussion 9B

1. Echoes in a Room: Canceling the Echo

Consider a simple acoustic model of a room that produces one echo. The echo arrives 3 samples after the original sound and has amplitude 0.6 relative to the direct sound.

Thus, the impulse response of the room is

$$h[n] = \delta[n] + 0.6\,\delta[n-3].$$

A short audio signal is given by

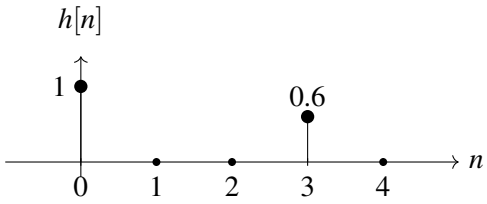
$$x[n] = \begin{cases} 1, & n = 0, \\ 0, & n = 1, \\ -1, & n = 2, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Plot the impulse response $h[n]$.

Answer: The impulse response is

$$h[n] = \delta[n] + 0.6\,\delta[n-3],$$

so it has impulses at $n = 0$ (amplitude 1) and $n = 3$ (amplitude 0.6), and is zero elsewhere.



(b) Write an expression for $y[n] = (x * h)[n]$ in terms of $x[n]$. Then compute $y[0], \ldots, y[6]$ for the short audio signal above.

Answer: Since $h[n] = \delta[n] + 0.6\,\delta[n-3]$, convolution simplifies to

$$y[n] = x[n] + 0.6\,x[n-3].$$

Adding the original signal and its delayed, scaled copy gives

$$y[n] = [1,\ 0,\ -1,\ 0.6,\ 0,\ -0.6,\ 0] \quad \text{for } n = 0 \ldots 6.$$

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(c) We now want to remove the echo from a recording $y[n] = x[n] + 0.6x[n-3]$, where $x[n]$ may be any signal that is zero for $n < 0$. Someone proposes subtracting a delayed, scaled copy of the recording:

$$z[n] = y[n] - 0.6y[n-3].$$

Express $z[n]$ in terms of $x[n]$. Does this recover the original signal exactly? Describe the remaining distortion.

Answer: Substituting the expressions for the recording and its delayed copy gives

$$\begin{aligned} z[n] &= \big(x[n] + 0.6x[n-3]\big) - 0.6\big(x[n-3] + 0.6x[n-6]\big) \\ &= x[n] - 0.36x[n-6]. \end{aligned}$$

The echo at delay three cancels, but the operation introduces a new echo at delay six with amplitude -0.36 relative to the original signal. This remaining echo is smaller and has its sign reversed, so the original signal is not recovered exactly in general.

(d) Instead, consider a system that uses its own previous output:

$$z[n] = y[n] + b\,z[n-3], \qquad z[n] = 0 \text{ for } n < 0.$$

Choose b so that $z[n] = x[n]$ for every input signal described in part (c). Explain why your choice removes the echo exactly.

Answer: If the desired output is $z[n] = x[n]$, substitution gives

$$x[n] = x[n] + 0.6x[n-3] + b\,x[n-3].$$

To satisfy this for every input, choose $b = -0.6$. To check that the resulting output really is $x[n]$, define the error $e[n] = z[n] - x[n]$. The recurrence becomes

$$e[n] = -0.6e[n-3].$$

The input and the cancellation system are initially at rest, so $e[n] = 0$ for $n < 0$. Every subsequent error sample is therefore also zero. The system subtracts a delayed copy of the recovered signal, rather than the echo-containing recording, so it does not introduce another echo.

2. Impulse Response of an IIR Filter

A system can feed a fraction of its previous output back into its current output, allowing the effect of a single input to persist. For example, if a microphone is brought too close to its speaker, sound that the microphone picks up will reverberate.

Consider the discrete-time LTI system

$$y[n] = \frac{1}{2}y[n-1] + x[n].$$

Assume the system starts at rest: $x[n] = y[n] = 0$ for $n < 0$.

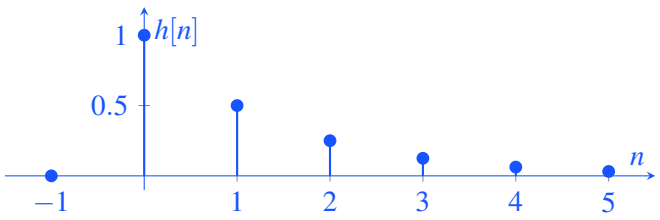
(a) Let $x[n] = \delta[n]$. Use the recurrence to compute $y[0], \ldots, y[5]$ and sketch these samples. Identify an expression for the impulse response $h[n]$ for all integers n.

Answer: Since $y[-1] = 0$ and $x[0] = 1$, we have $h[0] = 1$. For $n \geq 1$, the input is zero, so each output is half the previous output:

$$h[0], \ldots, h[5] = 1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}.$$

Thus,

$$h[n] = \begin{cases} \left(\frac{1}{2}\right)^n, & n \geq 0, \\ 0, & n < 0. \end{cases}$$



(b) Why does the output remain nonzero after the impulse has ended? Now consider replacing the coefficient $\frac{1}{2}$ by $\frac{3}{4}$, and then by $-\frac{1}{2}$. For each replacement, write the first four impulse-response samples and describe how the response differs from part (a). Does the original response ever become exactly zero? Explain why this is called an *infinite impulse response* (IIR) filter.

Answer: Even after the input becomes zero, the previous output contributes to the next output. The impulse's effect is therefore carried forward by the recurrence.

With coefficient $\frac{3}{4}$, the first four samples are $1, \frac{3}{4}, \frac{9}{16}, \frac{27}{64}$. They remain positive and decay more slowly than in part (a), since each step retains a larger fraction of the preceding output.

With coefficient $-\frac{1}{2}$, the first four samples are $1, -\frac{1}{2}, \frac{1}{4}, -\frac{1}{8}$. Their magnitudes are the same as in part (a), but their signs alternate.

The original response approaches zero but never equals zero at any finite $n \geq 0$. It has infinitely many nonzero samples, hence the name IIR. "Infinite" describes the duration of the response, not its amplitude.

(c) Return to the original coefficient $\frac{1}{2}$. Let $x[n] = 1$ for $n = 0,\ 1,\ 2$ and zero otherwise. Using the impulse response from part (a), compute the output through the convolution sum

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k].$$

Sketch $x[k]$ and $h[n-k]$ as functions of k for $n = 1$ and $n = 3$. Where do nonzero samples from each signal overlap? Then calculate $y[0], \ldots, y[5]$

Rewrite $y[n]$ as a sum of shifted copies of $h[n]$. Why does the output continue after the input ends?

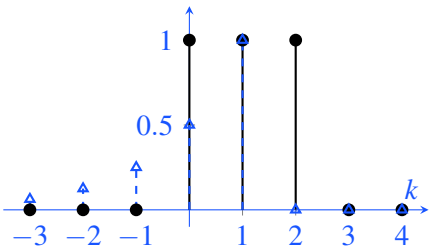
Answer: The input is nonzero only at $k = 0, 1, 2$. Also, $h[n-k]$ is nonzero only when $n - k \geq 0$, or $k \leq n$. The samples at times 0, 1, and 2 contribute $h[n]$, $h[n-1], and h[n-2]$, respectively, but each contribution begins only when its input sample arrives. At $n = 0$ only $x[0]$ contributes; at $n = 1$, $x[0]$ and $x[1]$ contribute; and for $n \geq 2$, all three contribute. Thus the contributing indices satisfy $0 \leq k \leq \min(2, n)$. For $n \geq 0$,

$$y[n] = \sum_{k=0}^{\min(2,n)} \left(\frac{1}{2}\right)^{n-k},$$

and $y[n] = 0$ for $n < 0$.

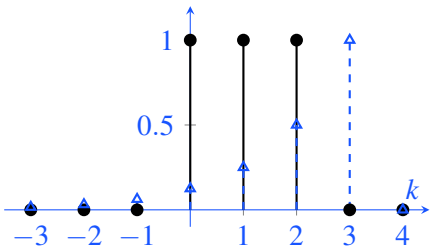
As a function of k, $h[n-k]$ is a flipped and shifted version of $h[k]$: it increases toward the value 1 at $k = n$, then is zero for $k > n$. The sketches below show part of its tail, which continues to the left. At $n = 1$, the nonzero samples overlap at $k = 0, 1$; at $n = 3$, they overlap at $k = 0, 1, 2$.

$n = 1$: overlap at $k = 0, 1$



$x[k]$ $h[1-k]$

$n = 3$: overlap at $k = 0, 1, 2$



$x[k]$ $h[3-k]$

Multiplying the overlapping samples and adding gives

$$y[0] = 1, \qquad y[3] = \frac{1}{8} + \frac{1}{4} + \frac{1}{2} = \frac{7}{8},$$
$$y[1] = \frac{1}{2} + 1 = \frac{3}{2}, \qquad y[4] = \frac{1}{16} + \frac{1}{8} + \frac{1}{4} = \frac{7}{16},$$
$$y[2] = \frac{1}{4} + \frac{1}{2} + 1 = \frac{7}{4}, \qquad y[5] = \frac{1}{32} + \frac{1}{16} + \frac{1}{8} = \frac{7}{32}.$$

Since $x[n] = \delta[n] + \delta[n-1] + \delta[n-2]$, the same output can be written as

$$y[n] = h[n] + h[n-1] + h[n-2].$$

Each of the three input samples produces a lasting impulse response. After the input ends at $n = 2$, these responses continue to contribute to the output. Their sum halves at each subsequent step, approaching zero without becoming exactly zero.

(d) **When is feedback stable?** A system is *bounded-input, bounded-output (BIBO) stable* if every bounded input produces a bounded output. You may use the following fact: a DT-LTI system is BIBO stable exactly when its impulse response is *absolutely summable*, meaning

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty.$$

Replace the coefficient $\frac{1}{2}$ in the original system by a real constant a:

$$y[n] = a\,y[n-1] + x[n],$$

with the system initially at rest. Find its impulse response $h_a[n]$ and determine exactly which values of a make the system BIBO stable. Justify your answer using the absolute sum.

Answer: As in part (a), the impulse gives $h_a[0] = 1$ and each subsequent sample is a times the previous one. Thus, for $a \neq 0$,

$$h_a[n] = \begin{cases} a^n, & n \geq 0, \\ 0, & n < 0. \end{cases}$$

For $a = 0$, $h_0[n] = \delta[n]$. In either case, the absolute sum is

$$\sum_{n=-\infty}^{\infty} |h_a[n]| = 1 + |a| + |a|^2 + \cdots.$$

If $|a| < 1$, this geometric series converges to $1/(1-|a|)$. If $|a| \geq 1$, its terms do not approach zero, so the series diverges. The system is therefore BIBO stable exactly when $-1 < a < 1$. In particular, alternating signs for negative a cannot make the *absolute* sum converge when $|a| \geq 1$.

(e) **A bounded response is not enough.** Set $a = -1$. A unit-step input produces the bounded output $1, 0, 1, 0, \ldots$, even though part (d) says this system is unstable. Find a bounded input, zero for $n < 0$, that produces an unbounded output. Give an expression for that output and explain why this demonstrates BIBO instability.

Answer: Choose an input whose signs alternate:

$$x[n] = \begin{cases} (-1)^n, & n \geq 0, \\ 0, & n < 0. \end{cases}$$

This input is bounded by 1 in magnitude. Using $h_{-1}[n] = (-1)^n$ for $n \geq 0$, convolution gives

$$y[n] = \sum_{k=0}^{n} (-1)^k (-1)^{n-k} = \sum_{k=0}^{n} (-1)^n = (n+1)(-1)^n, \qquad n \geq 0.$$

For any fixed n, all contributions have the same sign, so their magnitudes add rather than cancel. The output is $1, -2, 3, -4, \ldots$, whose magnitude grows without bound. BIBO stability requires a bounded output for *every* bounded input; this one counterexample establishes instability.